WESTERN MICHIGAN UNIVERSITY

CS 5610 - Advanced R for Data Science

ECOSYSTEM DATA MINING AND MAPPING

Instructor
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PROJECT OVERVIEW

To streamline the process of gathering data from diverse sources, including websites and APIs, with minimal manual intervention. To implement an ETL(Extract, Transform, Load) pipeline that cleans, standardizes, and stores data securely, meeting specific data volumes and latency. A fully functional pipeline that automates the collection, cleaning, transformation, and integration of data. Develop a user-friendly dashboard that presents the extracted insights in a easily interpretable manner.

INTRODUCTION

The data is of two types, events and opportunities. An event is a type of opportunity and those are the ones that are open to everyone without eligibility restrictions. Whereas, the opportunities are may have the eligibility requirements. Our team has worked on opportunities data mapping pipeline where we have to extract data from diverse sources, such as Gmail alerts, newsletters, social media platforms, RSS feeds etc. The data extracted from all these sources are to be processed, cleaned and stored in a database which serves as a central repository for all the collected data. For the data to be scalable, we can use the AWS RDS or Google cloud SQL.

OPPORTUNITY TYPES

1) Accelerator 2) Business advisor 3) Business consultant 4) Catalyst

5) Business or executive coach 6) Competition 7) Credits 8) Ecosystem

9) Fellow, paid or subsidized 10) Fellowship 11) Forgivable loan 12) Grants

13) In-kind services 14) Incubator 15) Interns, paid or subsidized

16) Legislative initiative 17) Mentorship 18) Paid or subsidized interns

19) Pro-Bono consulting 20) Pro-bono legal 21) Scholarships 22) Subsidy

23) Tax credit 24) Volunteer

METHODOLOGY

Phase 1: Requirements Gathering

- Conduct stakeholder interviews to understand specific needs and constraints.
- Define data sources, volume estimates, and specific latency targets.

Phase 2: System Design and Planning

- Architect the data pipeline and machine learning framework.
- Design the dashboard interface, focusing on usability and accessibility.

Phase 3: Development and Implementation

- Implement the data collection scripts and ETL processes.
- Develop and train machine learning models for recommendation and insight generation.
- Create a visualization dashboard integrating real-time data and insights.

Phase 4: Testing and Deployment

- Perform comprehensive testing, including unit, integration, and user acceptance tests.
- Deploy the system in a scalable cloud environment.

Phase 5: Monitoring, Feedback, and Iteration

- Establish monitoring for system performance and user engagement.
- Collect and analyze user feedback for continuous improvement.

OBJECTIVES

- Automate Data Collection
- Browser Extension
- Efficient Data Processing
- Insight Extraction Through Machine Learning
- Interactive Data Visualization
- Scalable and Secure Deployment
- Create a User-Friendly Report

EXPECTED OUTCOMES

- Operational Data Collection Pipeline
- Automated Data Collection System
- Browser Extension
- Robust ETL Pipeline
- Actionable Insights
- Interactive Dashboard
- High-Quality Dataset
- Insightful Data Analysis

DATA PIPELINE ENGINE FLOW DIAGRAM

Data Source Identification and Access

- Catalog primary data sources (websites, APIs) with access credentials.
- Validate data sources for reliability and update frequency.

Data Collection and Integration

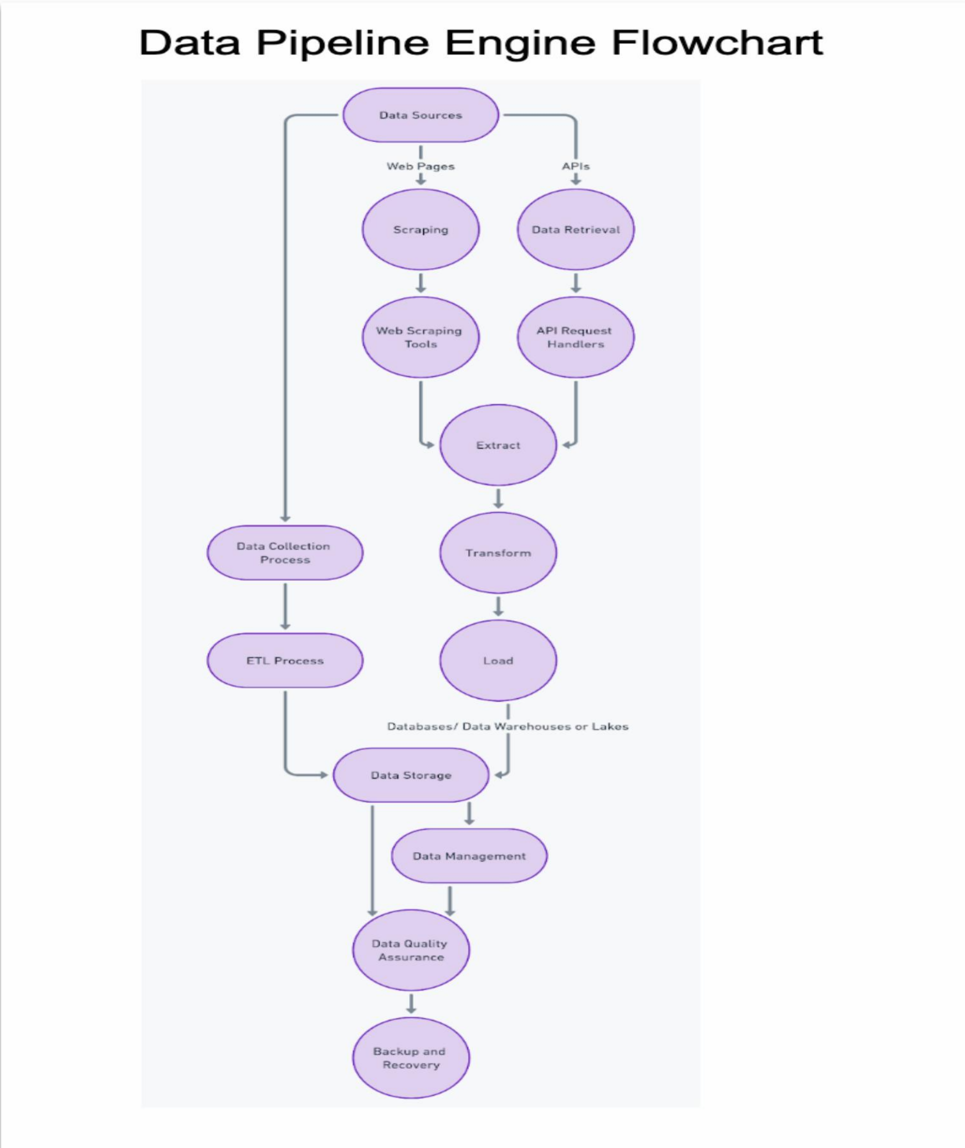
- Design and develop web scraping scripts with error handling and rate limiting.
- [Example sources: yawbyrd.com, incubatorlist.com]
- Configure API clients with authentication for dynamic data fetching.

ETL Pipeline Development

- Architect ETL workflow, identifying data transformation and normalization steps.
- Implement data validation checks to ensure quality and consistency.

Initial Data Cleaning and Preprocessing

- Standardize text data, including encoding and formatting.
- Identify and remove duplicate entries across data sources.



DATASET

- The dataset contains 151 entries and 8 columns.
- These columns include Title, State, Type, Description, Amount, Deadline, Website, and Date Posted.
- The Title column contains the name of the opportunity, such as grants, accelerators etc.
- The State column refers to the geographical location applicable for the opportunity, including "all US" and "varies".
- The Type column describes the type of opportunity, such as grant, accelerator etc.
- The Description column provides a brief description of the opportunity.
- The Amount column mentions the funding amount available, which appears to be in various formats.
- The Deadline column indicates the application deadline, which is in various formats.
- The Website column contains the URL for more information about the opportunity.
- Date Posted column refers to the date the opportunity was posted.

TECHNICAL REQUIREMENTS

- Programming Languages: R, Python
- Database: SQL
- OpenAI
- GitHub

R

R is an open-source programming language that is widely used as a statistical software and analysis tool. This generally comes with Command-line interface. R is available across widely used platforms like Windows, Linux, and macOS. Also, the R programming language is the latest cutting-edge tool.

PYTHON

Python is a high-level, interpreted programming language known for its simplicity and readability. It offers extensive libraries and frameworks that facilitate various tasks such as web development, data analysis, machine learning, and automation.

SQLite

SQLite is a lightweight, serverless, relational database management system. It is self-contained, meaning it requires minimal setup and administration. SQLite databases are stored as single files, making them easy to manage and transport. It is commonly used for embedded systems, mobile applications, and small to medium-scale web applications where a full-fledged DBMS may be overly complex.

GitHub

It is a code hosting platform for collaboration and version control. It is a website and cloud-based service that helps developers store and manage their code, as well as track and control changes to their code.

Data Collection Script:

- In our project, we utilized a Python script tailored to extract URLs from a designated Google Sheet, validate their authenticity, and subsequently store them within a SQLite database.
- The script's architecture incorporates several crucial steps for seamless execution.
- Initially, we imported essential libraries such as `gspread` for Google Sheets access, `requests` for HTTP requests, `BeautifulSoup` for web scraping, `sqlite3` for SQLite database interaction, and `urlparse` for URL parsing.
- Following library imports, the script initiated authentication procedures and established access to the specified Google Sheet.
- Subsequently, URLs were extracted from a designated column within the Google Sheet, with due diligence given to skipping the header row.
- The script then connected to a SQLite database named "scraped_data.db," where a cursor object facilitated SQL query execution.
- Upon database connection, a table named "scraped_urls" was created if absent, comprising two columns: `id` (primary key) and `url`.
- Iterating through the extracted URLs, the script meticulously validated each URL's scheme (`http` or `https`) and network location.

- Valid URLs absent from the database were inserted into the "scraped_urls" table, while duplicates and invalid URLs were disregarded.
- This Python script represents a pivotal component of our data collection pipeline, enabling the systematic extraction, validation, and storage of URLs for subsequent analysis and visualization within our ecosystem mapping platform.

Name	Type	Size	Value
client	client.Client	1	Client object of gspread.client module
c	Cursor	1	Cursor object of sqlite3 module
existing_row_count	int	1	1
scope	list	2	['https://spreadsheets.google.com/feeds', 'https://www.googleapis.com/ ...
urls	list	150	['https://aboutherculture.com/micro-grant-opportunities/', 'https://de ...
creds	oauth2.service_account.Credentials	1	Credentials object of google.oauth2.service_account module
parsed_url	parse.ParseResult	6	ParseResult object of urllib.parse module
sheet	spreadsheet.Spreadsheet	1	Spreadsheet object of gspread.spreadsheet module
key_file_path	str	17	Applications.json
sheet_id	str	44	1jFpMWLn_emhV0roxpVYU6_yGi7-KPNWgyikv_fy1dPY
url	str	152	https://www.seattle.gov/office-of-economic-development/small-business/ ...
worksheet	worksheet.Worksheet	1	Worksheet object of gspread.worksheet module

```
In [1]: runfile('/Users/rampal/Downloads/PROJECT R/NewAPIR.py', wdir='/Users/rampal/Downloads/PROJECT R')
Fetching URL: https://aboutherculture.com/micro-grant-opportunities/
URL already exists: https://aboutherculture.com/micro-grant-opportunities/
Fetching URL: https://developer.amazon.com/apps-and-games/small-business-program
URL already exists: https://developer.amazon.com/apps-and-games/small-business-program
Fetching URL: https://www.antler.co/platform/#apply-now
URL already exists: https://www.antler.co/platform/#apply-now
Fetching URL: https://www.passioncollective.co/atomic-grants
URL already exists: https://www.passioncollective.co/atomic-grants
Fetching URL: https://www.auroratechaward.com/
URL already exists: https://www.auroratechaward.com/
Fetching URL: https://www.awesomefoundation.org/en
URL already exists: https://www.awesomefoundation.org/en
Fetching URL: https://aws-startup-lofts.com/emea/program/accelerator
URL already exists: https://aws-startup-lofts.com/emea/program/accelerator
Fetching URL: https://becauseinternational.org/accelerator
URL already exists: https://becauseinternational.org/accelerator
Fetching URL: https://www.canva.com/canva-for-nonprofits/
URL already exists: https://www.canva.com/canva-for-nonprofits/
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URL already exists: https://mercatuscenter.formstack.com/forms/emergent_ventures
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URL already exists: https://www.f6s.com/programs
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URL already exists: https://fincaventures.awardsplatform.com/
Fetching URL: https://watson.is/ford-fund-fellowship/
URL already exists: https://watson.is/ford-fund-fellowship/
Fetching URL: https://gratitude-network.org/get-involved/apply-to-be-a-gratitude-leader/
URL already exists: https://gratitude-network.org/get-involved/apply-to-be-a-gratitude-leader/
Fetching URL: https://roddenberryfoundation.org/our-work/catalyst-fund/
URL already exists: https://roddenberryfoundation.org/our-work/catalyst-fund/
Fetching URL: https://savvyfellows.com/team/
URL already exists: https://savvyfellows.com/team/
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Fetching URL: https://www.wemakechange.org/
URL already exists: https://www.wemakechange.org/
Fetching URL:
Invalid URL:
Fetching URL: https://www.foundersfactory.africa/gen-f-eir-initiative
URL already exists: https://www.foundersfactory.africa/gen-f-eir-initiative
Fetching URL: https://www.grindstonexl.com/grindstone-accelerator
URL already exists: https://www.grindstonexl.com/grindstone-accelerator
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URL already exists: https://www.dmzlaunchpad.ca/courses/black-innovation-launchpad
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URL already exists: https://www.ic.gc.ca/eic/site/152.nsf/eng/00013.html
Fetching URL: https://www.toronto.ca/business-economy/business-operation-growth/business-incentives/commercial-space-rehabilitation-grant-program/
URL already exists: https://www.toronto.ca/business-economy/business-operation-growth/business-incentives/commercial-space-rehabilitation-grant-program/
Fetching URL: https://www.desjardins.com/ca/about-us/social-responsibility-cooperation/cooperative-movement/goodspark-grants/index.jsp
URL already exists: https://www.desjardins.com/ca/about-us/social-responsibility-cooperation/cooperative-movement/goodspark-grants/index.jsp
Fetching URL: https://impactfundingsolutions.com/impact-grant-application-for-businesses/
```
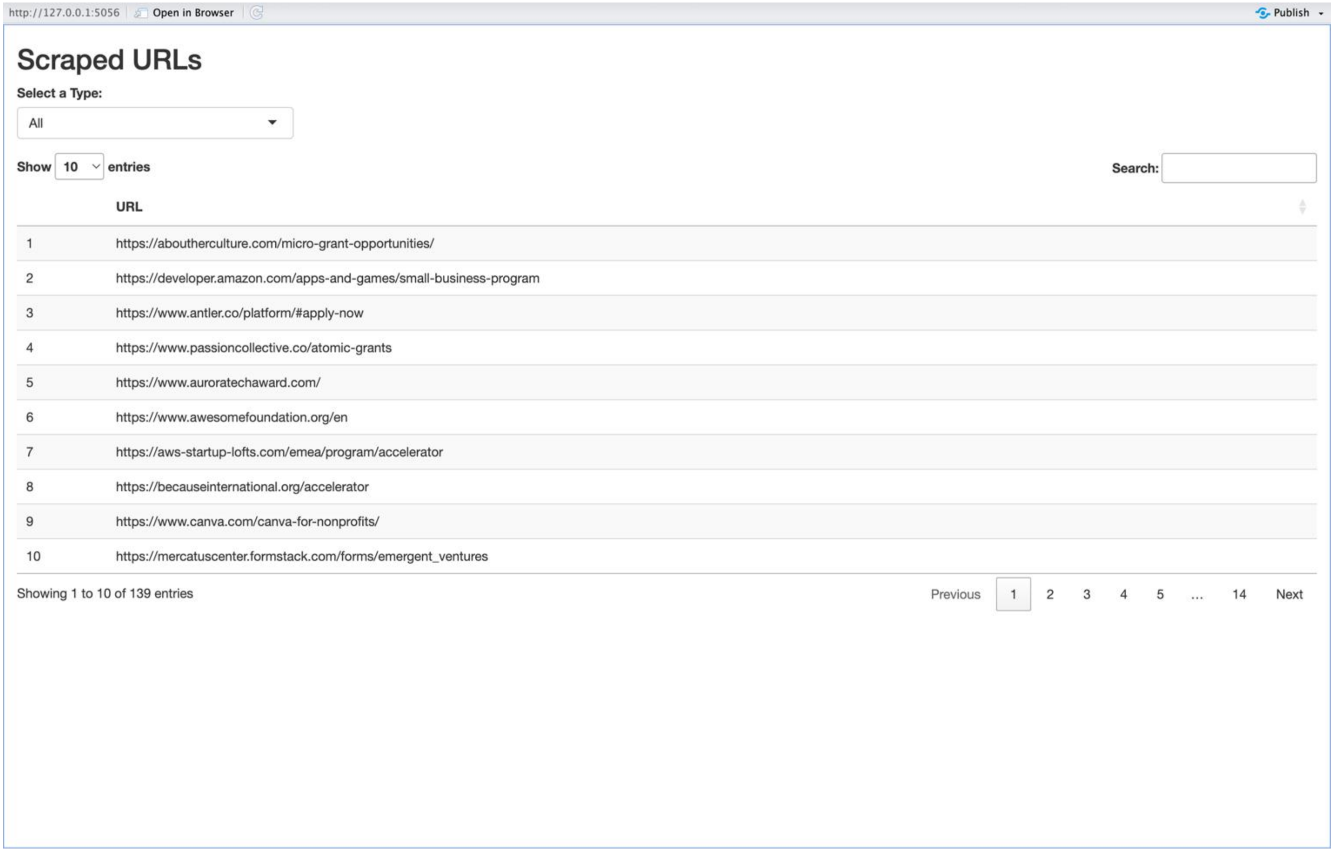
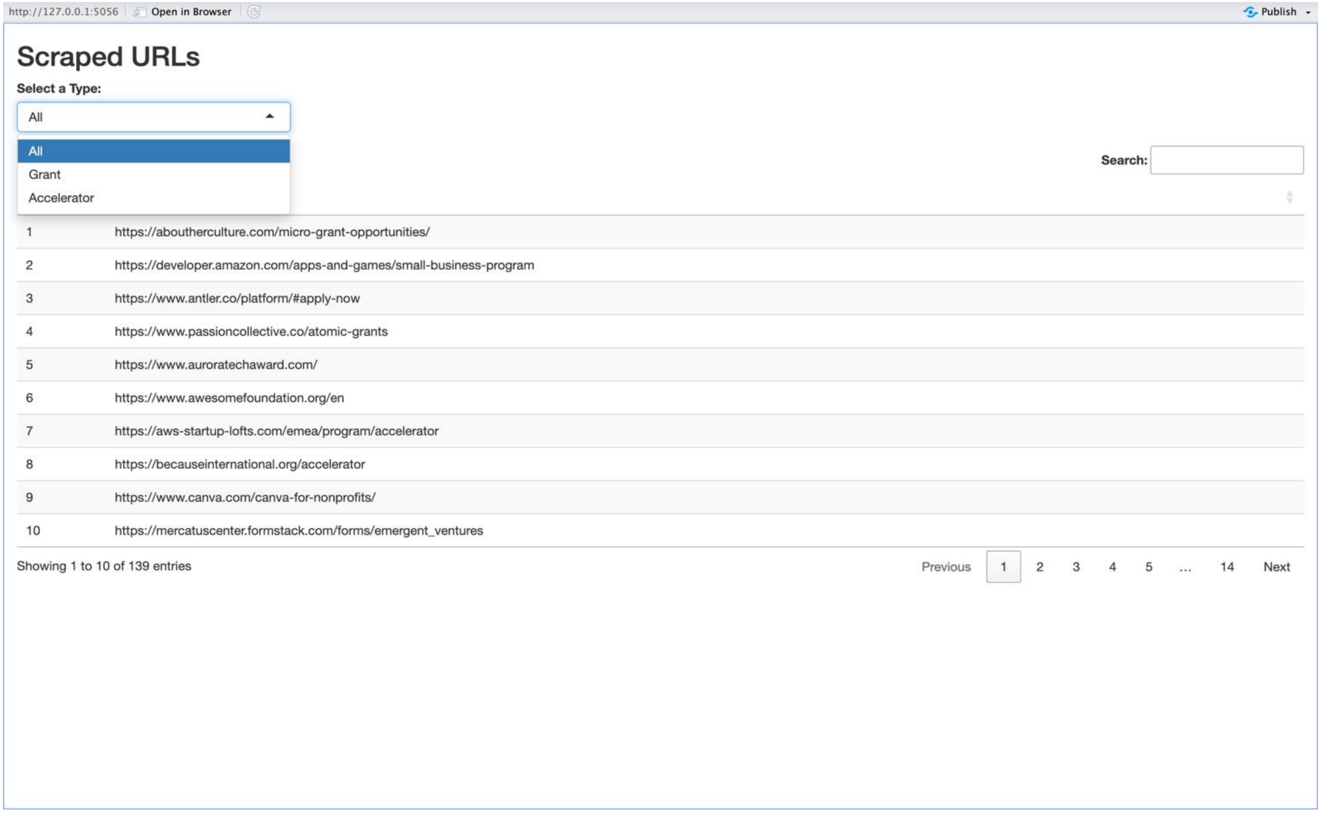

Integration of Shiny App:

In our project, the Shiny App serves as a user interface for viewing scraped URLs stored in a SQLite database, allowing users to filter URLs based on type selection and presenting them in a DataTable format. This code installs and loads necessary libraries, defines a user interface (UI) including a dropdown menu for selecting URL types and a DataTable for displaying URLs, establishes server logic to filter URLs based on type selection, render the table output, and close the database connection, and finally runs the Shiny application.




http://127.0.0.1:5056 Open in Browser Publish

Scraped URLs

Select a Type:

All

- All
- Grant
- Accelerator

Search:

1	https://aboutherculture.com/micro-grant-opportunities/
2	https://developer.amazon.com/apps-and-games/small-business-program
3	https://www.antler.co/platform/#apply-now
4	https://www.passioncollective.co/atomic-grants
5	https://www.auroratechaward.com/
6	https://www.awesomefoundation.org/en
7	https://aws-startup-lofts.com/emea/program/accelerator
8	https://becauseinternational.org/accelerator
9	https://www.canva.com/canva-for-nonprofits/
10	https://mercatuscenter.formstack.com/forms/emergent_ventures

Showing 1 to 10 of 139 entries

Previous 1 2 3 4 5 … 14 Next

CONCLUSION

We are able to retrieve the opportunity URLs from the Google spreadsheets provided into a database and created a RShiny app to visualize the data from the database. We have successfully integrated the python script that we developed with the RShiny app to priview the opportunites data.

FUTURE SCOPE

The future scope of the project would be to retieve the data from the Gmail alerts, RSS feeds, social media platforms etc., and perform necessary data cleansing and processing of the data and store the data in a database or datawarehouse lakes which would serve as the central repository for all the collected data.